A195194

Parity of an infinite nested reversion

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AI disclosure. ChatGPT (OpenAI) generated the mathematical arguments, Lean proofs, and this manuscript in response to Dev Goyal's requests. Dev Goyal is the sole human author. The source sequence and conjecture or observation are credited to Paul D. Hanna. **Independent human review is pending.**

Abstract

For arbitrary weights in a commutative ring, we construct a unique infinite family of normalized nested reversions and prove finite-depth coefficient stabilization. For weights $1, 2, 3, \ldots$, the first tail has odd coefficients exactly at powers of two. This is a statement about formal series, independent of analytic convergence.

1 Source, notation, and scope

The source is Paul D. Hanna's OEIS entry A195194 [1], retrieved on 24 October 2026. Its initial sequence index is 1. We also retain the formal zero constant coefficient at index zero.

All series are formal. The set $\mathbb{N}$ includes zero, and $[x^n]U$ denotes a coefficient. A normalized series is $x + O(x^2)$; $O(x^r)$ means divisibility by x^r, over any coefficient ring used below. We write $U \circ V = U(V(x))$ only when $V(0) = 0$. Compositional inversion, denoted $\mathrm{Rev}(U)$ or $U^{\langle -1 \rangle}$, is distinct from multiplicative inversion of a unit. Congruences are coefficientwise; ordinary products and compositional iterates are distinct.

2 Formal algebra used in the proof

Lemma 2.1 (Triangular construction)**.** *Let K be a commutative ring. Suppose Φ preserves normalized series and improves agreement by one degree:*

$$G - H = O(x^r) \quad \Longrightarrow \quad \Phi(G) - \Phi(H) = O(x^{r+1}) \qquad (r \geq 2).$$

Then Φ has a unique normalized fixed point. The same conclusion holds for a family of series if the degree-n equations, at every family index, determine the degree-n coefficients from strictly lower degrees.

Proof. The degree-n coefficient of $\Phi(G)$ depends only on coefficients of G below degree n. Fix the constant and linear coefficients, and then define every later coefficient successively. This gives a fixed point; the same induction proves uniqueness. For a family, perform the induction on degree simultaneously at every family index. The same induction applies with only the constant coefficient prescribed, provided agreement is improved starting at degree one. It also applies to an operator on all series when agreement is improved starting at degree zero. For the

operators used in this paper, the recurrence requires only ring operations, so it also constructs integral solutions when $K = \mathbb{Z}$. Equivalently, iterating Φ from any normalized initial series fixes one further coefficient at each step. □

Lemma 2.2 (Inversion and transport)**.** *A normalized series over any commutative ring has a unique normalized two-sided compositional inverse. A constant-one series has a unique multiplicative inverse with constant coefficient one. These inverses and valid composition commute with coefficient ring homomorphisms. Normalized compositional inversion preserves agreement modulo x^r.*

Proof. For normalized G, the degree-n equation in $G(H) = x$ determines h_n with multiplier one from earlier coefficients. Construct a right inverse J of H. Associativity gives $G = G \circ (H \circ J) = (G \circ H) \circ J = J$, so $H \circ G = x$ as well. The same associativity argument proves uniqueness. Multiplicative inversion follows similarly from the coefficient recurrence for $UV = 1$.

Each coefficient of a valid composition is a finite polynomial in the coefficients of its arguments. Ring homomorphisms preserve these polynomials and the defining equations, so uniqueness gives transport of inversion. If $G - H = O(x^r)$, composition with any normalized series preserves that agreement. Composing with the respective inverses then shows $\mathrm{Rev}(G) - \mathrm{Rev}(H) = O(x^r)$. □

Lemma 2.3 (Binary series and doubling)**.** *There is a unique normalized integral series $C = x + C^2$. Its reduction modulo two is*

$$P(x) = \sum_{k \geq 0} x^{2^k}.$$

For series U, V over $\mathbb{Z}/4\mathbb{Z}$, the equality $2U = 2V$ is equivalent to equality of their reductions modulo two.

Proof. The operator $G \mapsto x + G^2$ improves agreement because $G^2 - H^2 = (G - H)(G + H)$ and normalized $G + H$ has zero constant coefficient. In characteristic two, $U(x)^2 = U(x^2)$ for every formal series. Thus the normalized equation $U = x + U^2$ forces the coefficient at one to be one, every higher odd-index coefficient to be zero, and the coefficient at $2m$ to equal the coefficient at m. This gives exactly P.

For scalar residues modulo four, $2u = 2v$ holds precisely when u and v have the same parity. Apply this statement coefficientwise. In particular, this argument does not cancel a factor of two in a ring in which two is a zero divisor. □

3 A195194: infinite nested reversion

The title of A195194 is an infinite sequence of nested reversions [1]. It is useful to construct the more general family with arbitrary weights before specializing to $1, 2, 3, \ldots$.

Theorem 3.1 (Canonical nested tails)**.** *For every sequence $(w_j)_{j \geq 0}$ in a commutative ring K, there is a unique family of normalized series satisfying*

$$F_j = \mathrm{Rev}(x - w_j x F_{j+1}) \qquad (j \geq 0). \tag{1}$$

If the finite nests are defined by $F_j^{[0]} = x$ and

$$F_j^{[d+1]} = \mathrm{Rev}(x - w_j x F_{j+1}^{[d]}),$$

then $F_j^{[d]} - F_j = O(x^{d+2})$ for every $d, j \geq 0$.

Proof. The right inverse identity in (1) is equivalent to

$$F_j = x + w_j F_j \, F_{j+1}(F_j). \tag{2}$$

Both factors in the nontrivial term start in degree one. Consequently its degree-n coefficient depends only on coefficients below degree n of F_j and F_{j+1}. Lemma 2.1 constructs the unique family over K. No boundary condition at infinity is assumed. Since $x - w_j x F_{j+1}$ is normalized, Lemma 2.2 turns the right inverse equation into both inverse identities, proving (1).

At depth zero both the approximation and the true tail start with x, so they agree modulo x^2. Multiplication by x improves the agreement of their next tails by one degree, and compositional inversion preserves that improved agreement. Induction on d gives the stated bound. Thus the finite nested expressions agree with the canonical family through degree $d + 1$; their formal infinite limit is unambiguous. □

Theorem 3.2 (A195194 parity)**.** *Let $w_j = j + 1$, and let $A = F_0$ in Theorem 3.1, over $\mathbb{Z}$. For every $n \in \mathbb{N}$,*

$$[x^n]A \text{ is odd} \quad \Longleftrightarrow \quad n = 2^k \text{ for some } k \in \mathbb{N}.$$

Proof. Map the family to $\mathbb{F}_2[[x]]$ and write a bar for reduction. The second weight is two, so (2) gives $\overline{F}_1 = x$. The first weight is one, and therefore

$$\overline{A} = x + \overline{A}\,\overline{F}_1(\overline{A}) = x + \overline{A}^2.$$

Lemma 2.3 identifies $\overline{A}$ with P. The conclusion also holds at $n = 0$, where the coefficient is zero and the index is not a power of two. □

The formal implementation constructs the simultaneous family using the coefficient ring $\mathbb{N} \to K$, with its shift endomorphism. Evaluation at an index recovers (2). The theorem `source_existence_uniqueness` states the general source result using both inverse identities; `approximation_agrees` supplies the finite-depth bound, and `a195194_parity` gives the infinite parity statement. All are in `Round4.NestedReversion`.

4 Separate Lean proof and checked scope

The entry file is `lean/Entries/A195194.lean`. Its explicit theorem statements and proof terms use the namespace `FinalCollection.A195194`. The complete imported implementations and their shared mathematical dependencies are included in the collection. The following local theorem names give the proof map:

- `normalized_source`.
- `all_tails_normalized`.
- `all_tails_source`.
- `nested_definition_unique`.
- `first_tail_reversion`.
- `finite_nest_coefficients`.
- `binary_equation`.
- `exact_parity`.
- `arbitrary_weights_unique`.
- `arbitrary_finite_depth`.

The main implementation module is `Round4.NestedReversion`. The catalogue records the full original endpoint names and the new entry theorems. The project pins Lean 4.19.0 and Mathlib revision `c44e0c8ee63ca166450922a373c7409c5d26b00b`. The collection report records source recompilation and transitive axiom checks. Accepted axioms are propositional extensionality, classical choice, and quotient soundness. All index conclusions above are unbounded; exact finite coefficient checks in the package provide separate corroboration.

Limits and status. Convergence means coefficient stabilization of formal series, with no analytic assertion. The source starts at n=1. The manuscript consolidates completed results; it does not assert global novelty, exclusive priority, peer review, OEIS acceptance, or a new submission. Standard algebraic ingredients are not claimed as new discoveries.

References

[1] Paul D. Hanna, *OEIS A195194*, 11 September 2011. `https://oeis.org/A195194`. Retrieved 24 October 2026.